

Bangladesh University of Business & Technology (BUBT)



LAB Report

Course Code : CSE 210
Course Title : Operating Systems Lab
Experiment No. : 02

Name: Md. Fahim Islam
ID: 20245103035
Intake: 53
Section: 01
Program: B.Sc. Engg. in CSE

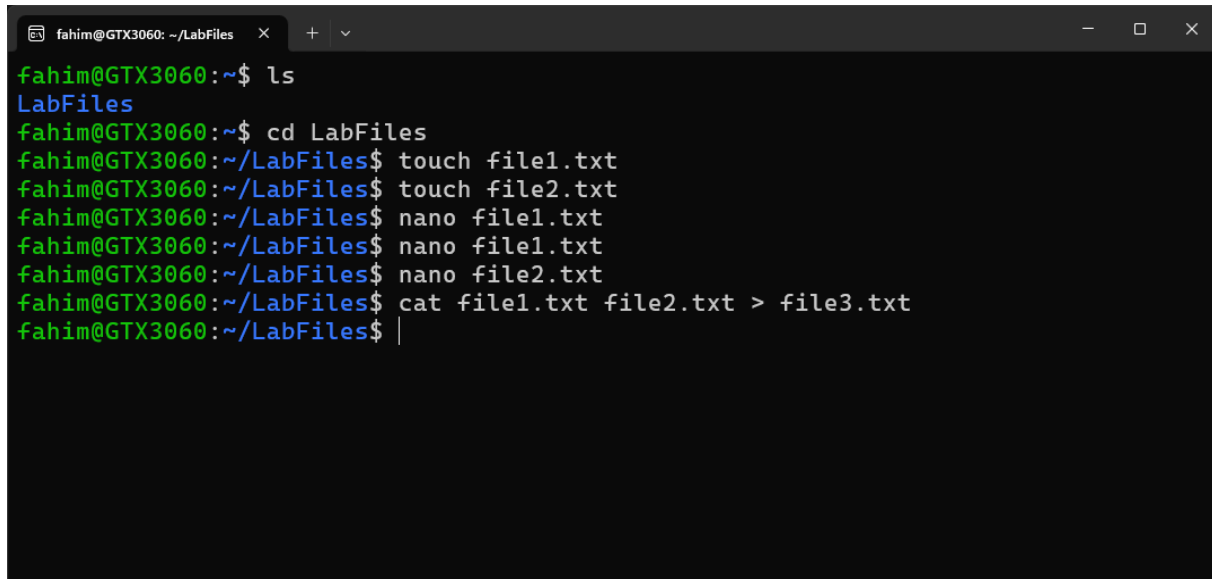
Name: Mishal Al Rahman
Dept. of Computer Science and
Engineering (CSE)
Bangladesh University of Business and
Technology

Date of Submission: 04/08/2025

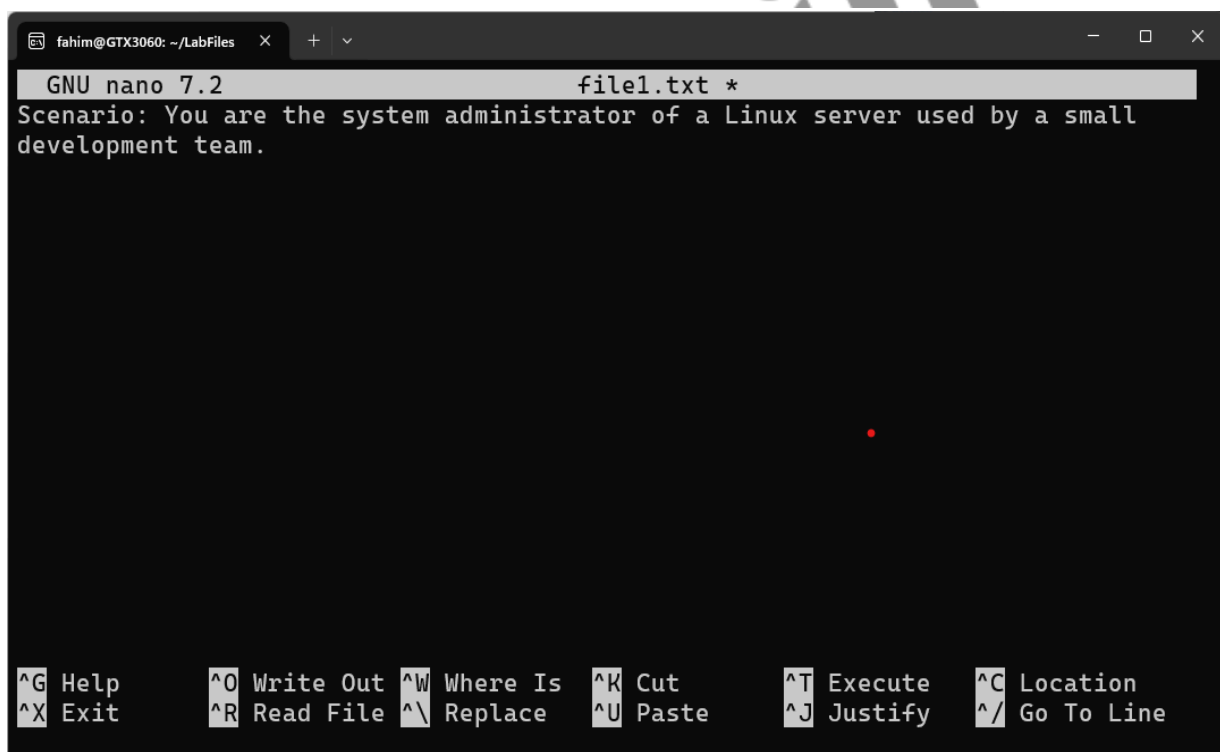
Signature of Teacher

1. Concatenate file1.txt and file2.txt to make a file3.txt with all the contents of file1.txt and file2.txt.

```
touch file1.txt
touch file2.txt
nano file1.txt
nano file2.txt
cat file1.txt file2.txt > file3.txt
```



```
fahim@GTX3060: ~/LabFiles
fahim@GTX3060:~$ ls
LabFiles
fahim@GTX3060:~$ cd LabFiles
fahim@GTX3060:~/LabFiles$ touch file1.txt
fahim@GTX3060:~/LabFiles$ touch file2.txt
fahim@GTX3060:~/LabFiles$ nano file1.txt
fahim@GTX3060:~/LabFiles$ nano file1.txt
fahim@GTX3060:~/LabFiles$ nano file2.txt
fahim@GTX3060:~/LabFiles$ cat file1.txt file2.txt > file3.txt
fahim@GTX3060:~/LabFiles$
```



```
GNU nano 7.2 file1.txt *
Scenario: You are the system administrator of a Linux server used by a small
development team.

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify  ^_ Go To Line
```

```
fahim@GTX3060: ~/LabFiles
GNU nano 7.2 file2.txt *
Recently, there have been some issues reported by team members, and you need
to troubleshoot and perform user management tasks.

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste       ^J Justify    ^/ Go To Line
```

```
fahim@GTX3060: ~/LabFiles
GNU nano 7.2 file3.txt
Scenario: You are the system administrator of a Linux server used by a small
development team.
Recently, there have been some issues reported by team members, and you need
to troubleshoot and perform user management tasks.

[ Read 4 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste       ^J Justify    ^/ Go To Line
```

2. Show all your running process list

ps aux

```
fahim@GTX3060: ~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.1  21796 12384 ?        Ss   11:33   0:00 /sbin/init
root         2  0.0  0.0   3060  1664 ?        Sl   11:33   0:00 /init
root         7  0.0  0.0   3076  1792 ?        Sl   11:33   0:00 plan9 --contr
root        53  0.0  0.1  42236 15360 ?        S<s  11:33   0:00 /usr/lib/syst
root        98  0.0  0.0  25004  6144 ?        Ss   11:33   0:00 /usr/lib/syst
systemd+  162  0.0  0.1  21452 12672 ?        Ss   11:33   0:00 /usr/lib/syst
systemd+  173  0.0  0.0  91020  7680 ?        Ssl  11:33   0:00 /usr/lib/syst
root       182  0.0  0.0   4236  2432 ?        Ss   11:33   0:00 /usr/sbin/cro
message+  183  0.0  0.0   9532  4864 ?        Ss   11:33   0:00 @dbus-daemon
root       200  0.0  0.1  17964  8448 ?        Ss   11:33   0:00 /usr/lib/syst
root       204  0.0  0.1 1756096 11776 ?        Ssl  11:33   0:00 /usr/libexec/
syslog    229  0.0  0.0 222508  5248 ?        Ssl  11:33   0:00 /usr/sbin/rsy
root      240  0.0  0.0   3160  1920 hvc0     Ss+  11:33   0:00 /sbin/agetty
root      249  0.0  0.0   3116  1792 tty1     Ss+  11:33   0:00 /sbin/agetty
root      265  0.0  0.2 107012 22396 ?        Ssl  11:33   0:00 /usr/bin/pyth
root      362  0.0  0.0   3064   896 ?        Ss   11:33   0:00 /init
root      363  0.0  0.0   3080  1024 ?        S    11:33   0:00 /init
fahim     364  0.0  0.0   6072  4864 pts/0    Ss   11:33   0:00 -bash
root      365  0.0  0.0   6820  4224 pts/1    Ss   11:33   0:00 /bin/login -f
fahim     456  0.0  0.1  20292 11264 ?        Ss   11:33   0:00 /usr/lib/syst
```

```
fahim@GTX3060: ~$ ps aux
root        53  0.0  0.1  42236 15360 ?        S<s  11:33   0:00 /usr/lib/syst
root        98  0.0  0.0  25004  6144 ?        Ss   11:33   0:00 /usr/lib/syst
systemd+  162  0.0  0.1  21452 12672 ?        Ss   11:33   0:00 /usr/lib/syst
systemd+  173  0.0  0.0  91020  7680 ?        Ssl  11:33   0:00 /usr/lib/syst
root       182  0.0  0.0   4236  2432 ?        Ss   11:33   0:00 /usr/sbin/cro
message+  183  0.0  0.0   9532  4864 ?        Ss   11:33   0:00 @dbus-daemon
root       200  0.0  0.1  17964  8448 ?        Ss   11:33   0:00 /usr/lib/syst
root       204  0.0  0.1 1756096 11776 ?        Ssl  11:33   0:00 /usr/libexec/
syslog    229  0.0  0.0 222508  5248 ?        Ssl  11:33   0:00 /usr/sbin/rsy
root      240  0.0  0.0   3160  1920 hvc0     Ss+  11:33   0:00 /sbin/agetty
root      249  0.0  0.0   3116  1792 tty1     Ss+  11:33   0:00 /sbin/agetty
root      265  0.0  0.2 107012 22396 ?        Ssl  11:33   0:00 /usr/bin/pyth
root      362  0.0  0.0   3064   896 ?        Ss   11:33   0:00 /init
root      363  0.0  0.0   3080  1024 ?        S    11:33   0:00 /init
fahim     364  0.0  0.0   6072  4864 pts/0    Ss   11:33   0:00 -bash
root      365  0.0  0.0   6820  4224 pts/1    Ss   11:33   0:00 /bin/login -f
fahim     456  0.0  0.1  20292 11264 ?        Ss   11:33   0:00 /usr/lib/syst
fahim     457  0.0  0.0  21148  3520 ?        S    11:33   0:00 (sd-pam)
fahim     501  0.0  0.0   6056  5120 pts/1    S+   11:33   0:00 -bash
fahim     520  0.0  0.0   5084  2560 pts/0    T    11:33   0:00 ping google.c
fahim     528  0.0  0.0   8280  4096 pts/0    R+   11:37   0:00 ps aux
fahim@GTX3060: ~$
```

3. ping google.com

ping google.com

```
fahim@GTX3060: ~$ ping google.com
PING google.com (172.217.167.206) 56(84) bytes of data.
64 bytes from del03s18-in-f14.1e100.net (172.217.167.206): icmp_seq=1 ttl=116 time=45.4 ms
64 bytes from del03s18-in-f14.1e100.net (172.217.167.206): icmp_seq=2 ttl=116 time=45.3 ms
64 bytes from del03s18-in-f14.1e100.net (172.217.167.206): icmp_seq=3 ttl=116 time=45.1 ms
64 bytes from del03s18-in-f14.1e100.net (172.217.167.206): icmp_seq=4 ttl=116 time=45.6 ms
^Z
[1]+  Stopped                  ping google.com
fahim@GTX3060:~$ |
```

4.zip file using tar command.

tar -czvf zip_files.tar.zip file1.txt file2.txt

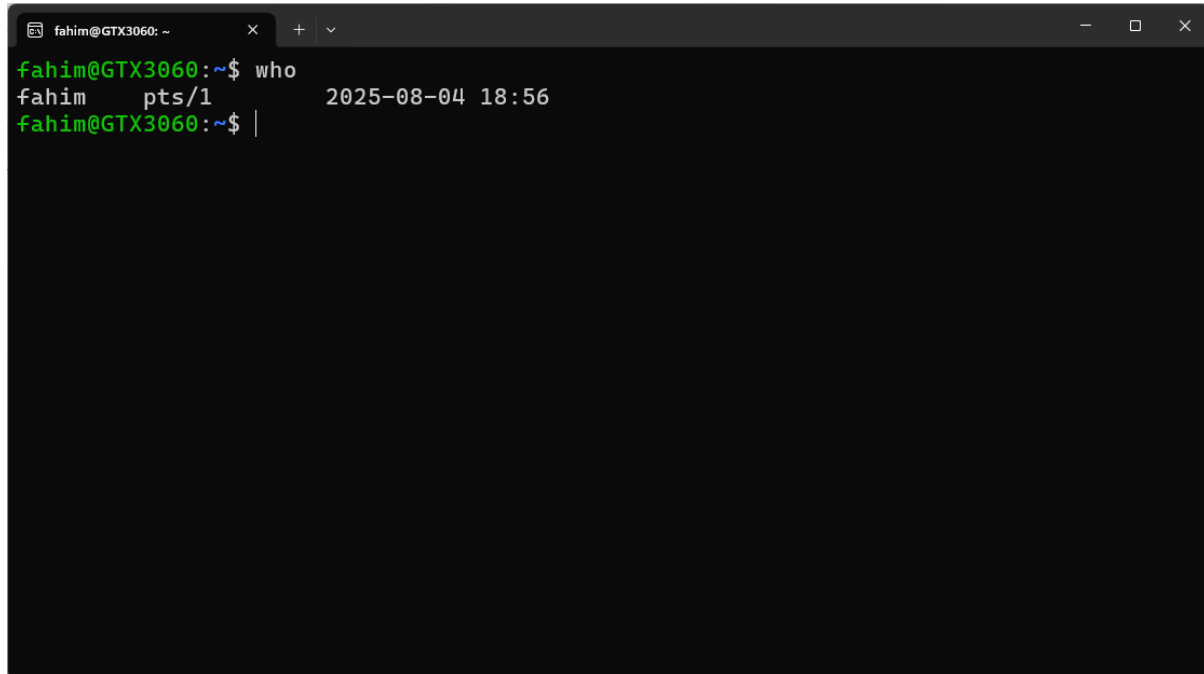
```
fahim@GTX3060: ~/LabFiles$ tar -czvf zip_files.tar.zip file1.txt file2.txt
file1.txt
file2.txt
fahim@GTX3060:~/LabFiles$
```

file1.txt	04/08/2025 11:45 AM	Text Document	1 KB
file2.txt	04/08/2025 11:47 AM	Text Document	1 KB
file3.txt	04/08/2025 11:47 AM	Text Document	1 KB
zip_files.tar.zip	04/08/2025 11:54 AM	WinRAR ZIP archive	1 KB

5. Troubleshooting:

5.1 Use the who command to identify the currently logged-in users on the system.

who

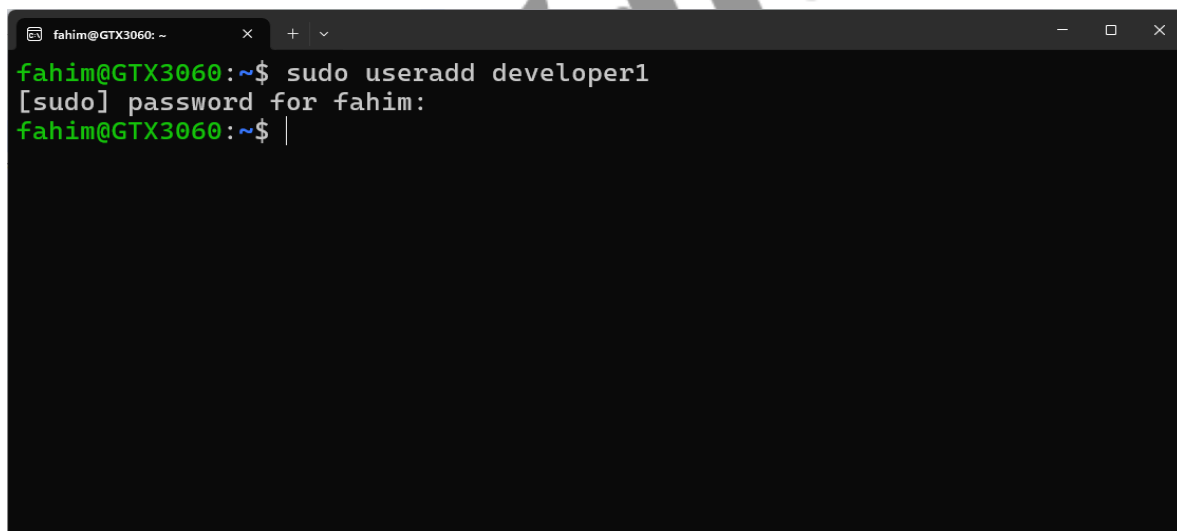
A terminal window titled 'fahim@GTX3060: ~' with standard window controls. The terminal shows the command 'who' being executed. The output is 'fahim pts/1 2025-08-04 18:56'. The prompt changes from 'fahim@GTX3060:~\$' to 'fahim@GTX3060:~\$' after the command is run.

```
fahim@GTX3060:~$ who
fahim pts/1 2025-08-04 18:56
fahim@GTX3060:~$ |
```

6. User Management:

6.1 Create a new user named developer1 using the useradd command.

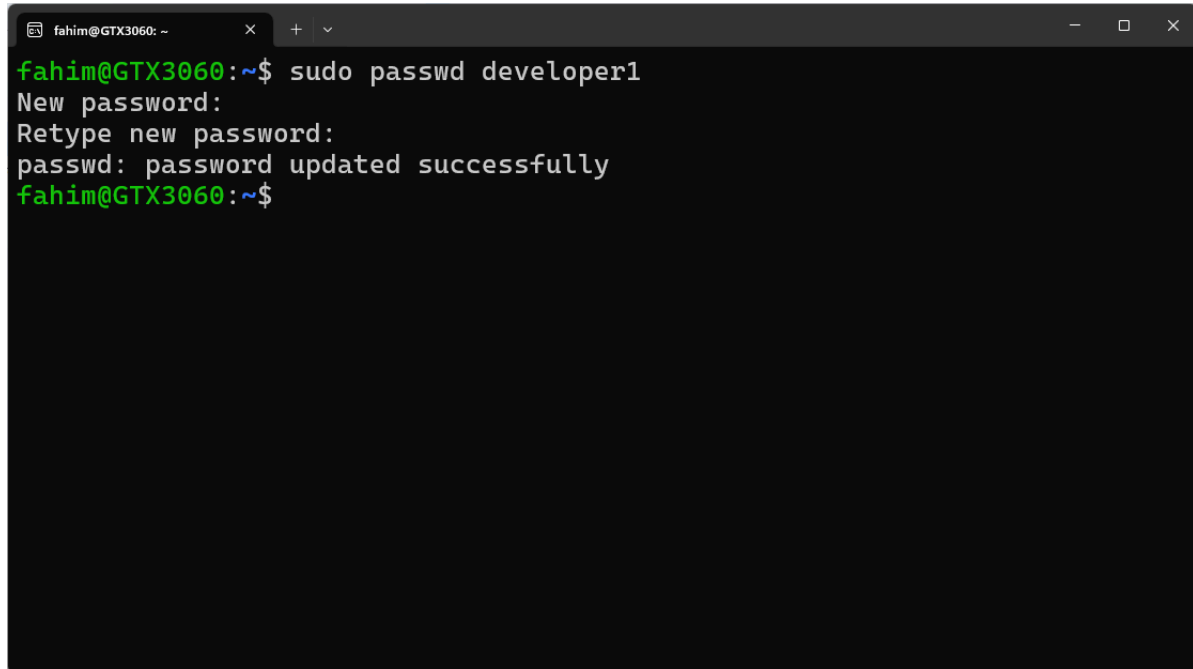
sudo useradd developer1

A terminal window titled 'fahim@GTX3060: ~' with standard window controls. The terminal shows the command 'sudo useradd developer1' being executed. It prompts for a password: '[sudo] password for fahim:'. The prompt returns to 'fahim@GTX3060:~\$' after the command is run.

```
fahim@GTX3060:~$ sudo useradd developer1
[sudo] password for fahim:
fahim@GTX3060:~$ |
```

6.2 Set a password for the newly created user using the passwd command.

`sudo passwd developer1`



```
fahim@GTX3060: ~$ sudo passwd developer1
New password:
Retype new password:
passwd: password updated successfully
fahim@GTX3060: ~$
```

6.3 Create a new group named development using the groupadd command.

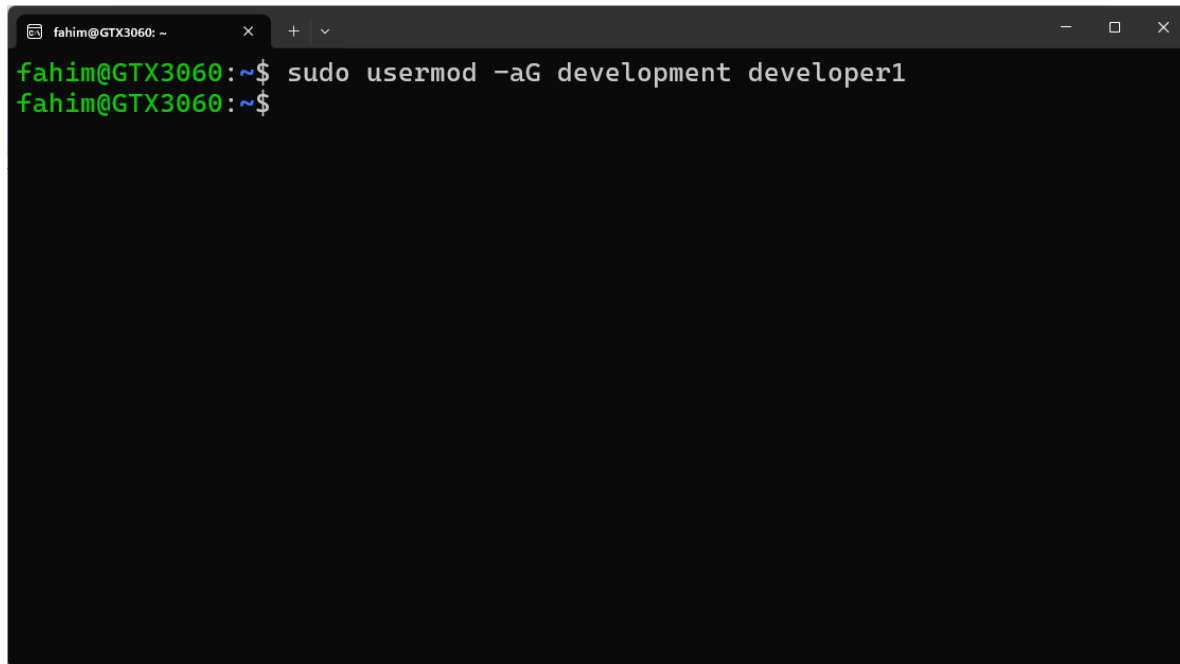
`sudo groupadd development`



```
fahim@GTX3060: ~$ sudo groupadd development
fahim@GTX3060: ~$
```

6.4 Add the user developer1 to the development group using the usermod command.

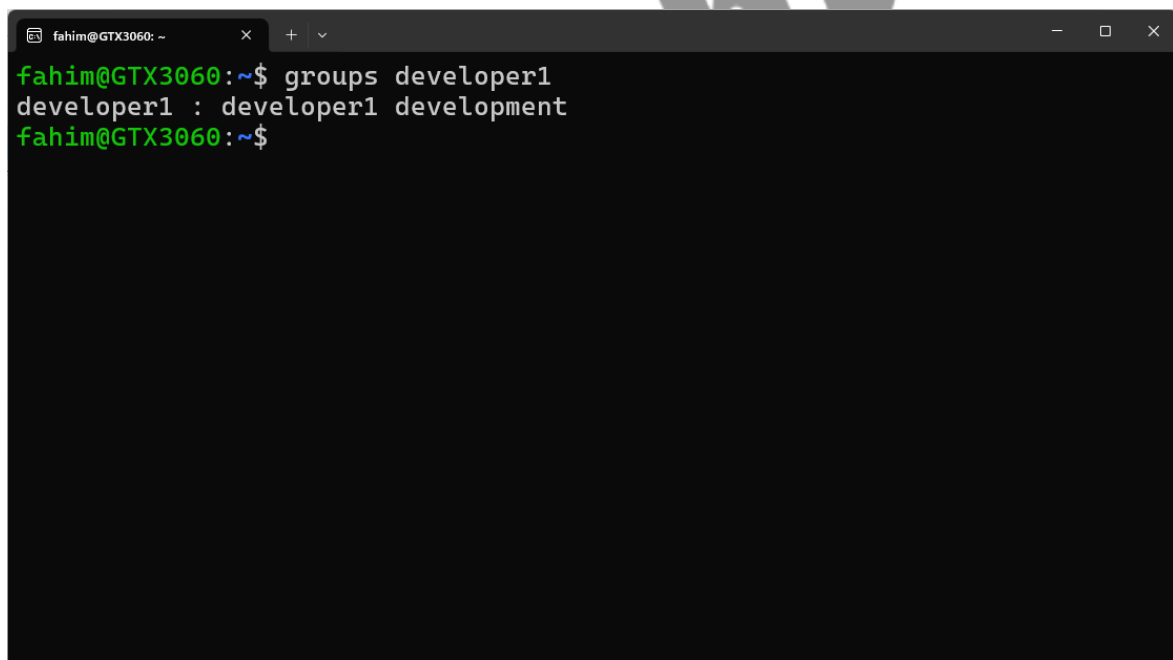
```
sudo usermod -aG development developer1
```

A terminal window with a dark background and light green text. The window title is 'fahim@GTX3060: ~'. The prompt is 'fahim@GTX3060:~\$'. The command 'sudo usermod -aG development developer1' has been entered and executed. The prompt returns to 'fahim@GTX3060:~\$' on the next line.

```
fahim@GTX3060:~$ sudo usermod -aG development developer1
fahim@GTX3060:~$
```

6.5 Check and display the group memberships of the user developer1 using the groups command.

```
groups developer1
```

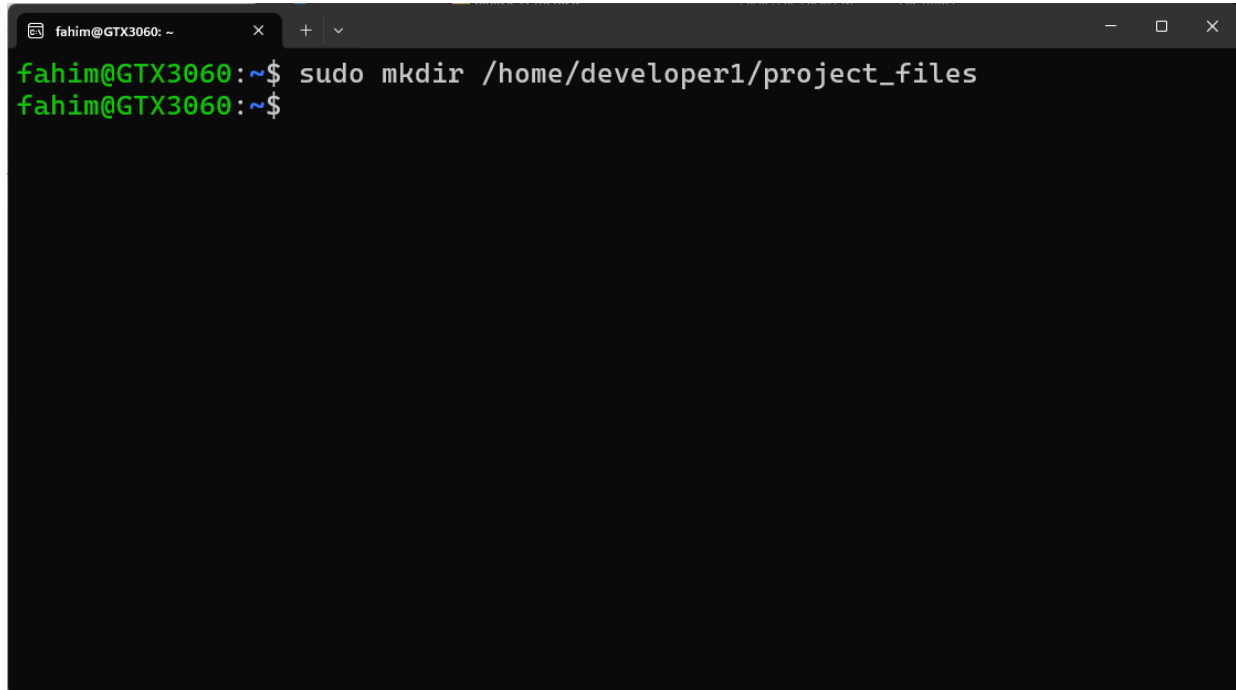
A terminal window with a dark background and light green text. The window title is 'fahim@GTX3060: ~'. The prompt is 'fahim@GTX3060:~\$'. The command 'groups developer1' has been entered and executed. The output is 'developer1 : developer1 development'. The prompt returns to 'fahim@GTX3060:~\$' on the next line.

```
fahim@GTX3060:~$ groups developer1
developer1 : developer1 development
fahim@GTX3060:~$
```


7. File Permission Adjustment:

7.1 Create a directory named `project_files` in the home directory of `developer1` using the `mkdir` command.

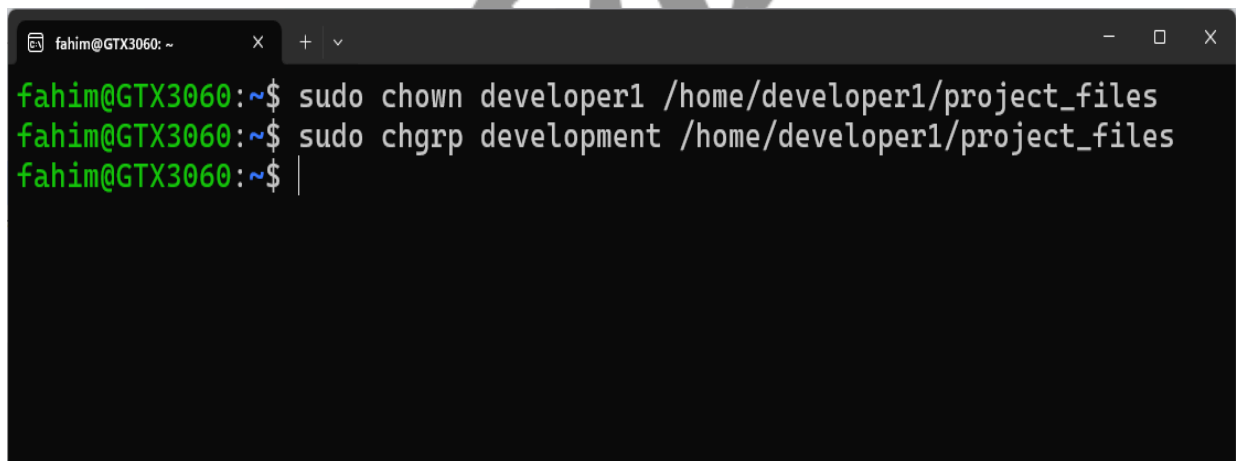
```
sudo mkdir /home/developer1/project_files
```

A terminal window titled 'fahim@GTX3060: ~' with a dark background. The prompt is 'fahim@GTX3060:~\$'. The command 'sudo mkdir /home/developer1/project_files' has been entered and executed, resulting in a new prompt 'fahim@GTX3060:~\$' on the next line.

```
fahim@GTX3060:~$ sudo mkdir /home/developer1/project_files
fahim@GTX3060:~$
```

7.2 Change the ownership of the `project_files` directory to `developer1` and the group to `development` using the `chown` and `chgrp` commands.

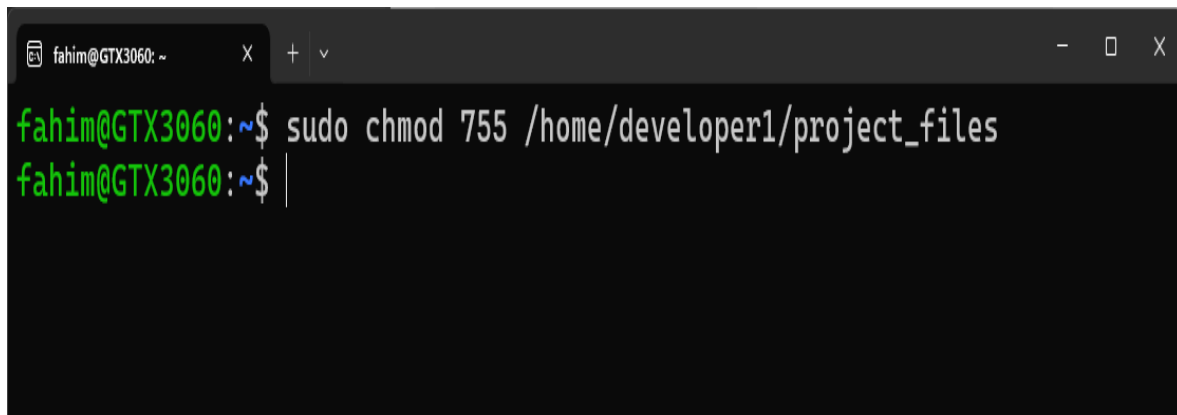
```
sudo chown developer1 /home/developer1/project_files
sudo chgrp development /home/developer1/project_files
```

A terminal window titled 'fahim@GTX3060: ~' with a dark background. The prompt is 'fahim@GTX3060:~\$'. Two commands are entered and executed: 'sudo chown developer1 /home/developer1/project_files' and 'sudo chgrp development /home/developer1/project_files'. The prompt returns after each command.

```
fahim@GTX3060:~$ sudo chown developer1 /home/developer1/project_files
fahim@GTX3060:~$ sudo chgrp development /home/developer1/project_files
fahim@GTX3060:~$
```

7.3 Ensure that only the owner (developer1) has write permissions on the project_files directory.

```
sudo chmod 755 /home/developer1/project_files
```



```
fahim@GTX3060: ~$ sudo chmod 755 /home/developer1/project_files
fahim@GTX3060: ~$
```

8. Conclusion:

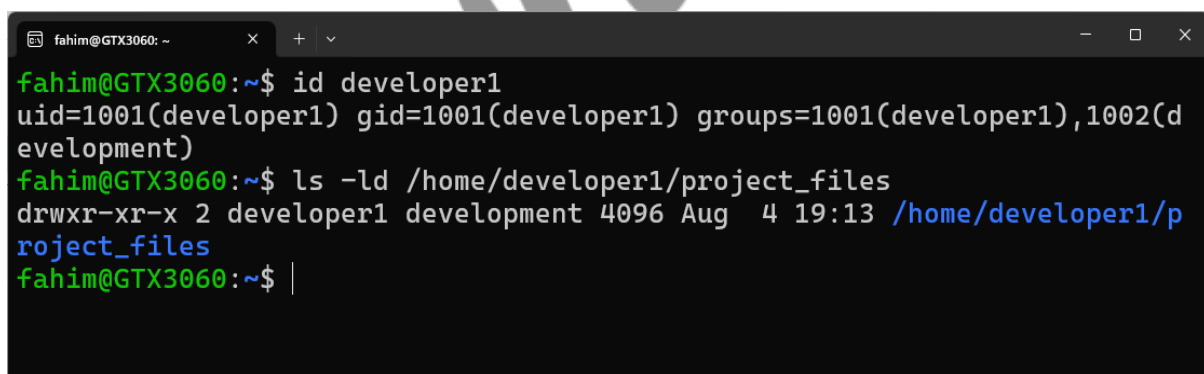
8.1 Summarize your troubleshooting findings and the actions taken to resolve any issues.

Had to use command "sudo mkdir /home/developer1" first because the using "sudo mkdir /home/developer1/project_files" this pops up
"No such file or directory."

8.2 Confirm that the new user developer1 has been successfully created, added to the development group, and that file permissions are set correctly.

```
id developer1
```

```
ls -ld /home/developer1/project_files
```



```
fahim@GTX3060: ~$ id developer1
uid=1001(developer1) gid=1001(developer1) groups=1001(developer1),1002(development)
fahim@GTX3060: ~$ ls -ld /home/developer1/project_files
drwxr-xr-x 2 developer1 development 4096 Aug  4 19:13 /home/developer1/project_files
fahim@GTX3060: ~$
```